2. Comparison of Software Development Methodologies

Four software development methodologies were considered for RP-Cloud Labs.

Methodology	Advantages	Disadvantages	Suitability
Waterfall	Clear phases, documentation, and predictable planning.	Changes are difficult and costly after a phase is completed.	Less suitable because requirements may change as colleges join.
Agile	Develops software in small increments, supports frequent feedback, and adapts to changing requirements.	Requires regular communication, collaboration, and active stakeholder participation.	Highly suitable because the system will evolve as more colleges join.
Prototyping	Produces an early model that helps users understand and improve requirements.	Repeated changes to the prototype may increase time and cost.	Useful for testing ideas about the user interface and session-request process.
Spiral	Focuses on risk analysis and repeated development cycles.	Can be expensive and complex to manage.	Useful for a large, high-risk system, but may be too complex for this small module feature.